Full Stack Assignment – 31

Assignment ID: FSD-31

Title: Certification Approval Management System

Duration: 3 Hours

1 Objective

Design and develop a **Certification Approval Management System** that allows applicants to submit certification requests and verifiers to validate them.
The system must enforce single-verification rules and ensure **data persistence using a database**.

2 User Roles

APPLICANT

- Register and log in to the system
- Submit certification requests
- View certification request status

VERIFIER

- Register and log in to the system
- View all certification requests
- Verify or reject certification requests

3 Functional Requirements

Authentication & Authorization

- Implement user registration and login
- Use **JWT-based authentication**
- Protect all backend APIs using JWT
- Enforce **role-based access control** on APIs and frontend views

Certification Management

- APPLICANT can submit certification requests with required details
- Certification requests must be stored in the database
- VERIFIER can verify or reject certification requests
- Certification status must be retrieved dynamically

4️⃣ Business Rule (Critical)

⚠️ Each certification request can be verified only once.

Once a request is verified or rejected, no further action must be allowed on it.

5️⃣ Technical Constraints (Mandatory)

- All certification and verification data **must persist in a database**
- Hardcoded or in-memory storage is **not allowed**
- Backend APIs must reflect actual database state
- JWT authentication and role validation are mandatory
- Use environment variables for sensitive configuration

6️⃣ Frontend Requirements

- Login and registration pages
- Role-based dashboards for APPLICANT and VERIFIER
- Certification request submission view for APPLICANT
- Certification review and verification view for VERIFIER

UI design is not evaluated; state enforcement and correctness matter.

7️⃣ Submission Requirements (Strict)

GitHub Repository

- Backend source code
- Frontend source code
- Proper folder structure
- Clear and meaningful commit history

Live Deployment

- Live frontend deployment link
- Live backend deployment link
- Backend must be connected to the database

README.md (Mandatory)

The README file must include:

- Project overview
- Tech stack used
- User roles and permissions
- API endpoints
- Database schema or collections

- Steps to run the project locally
- Live frontend and backend links

8️⃣ Evaluation Note

Evaluation focuses on **single-action enforcement, role control, and end-to-end execution**.